

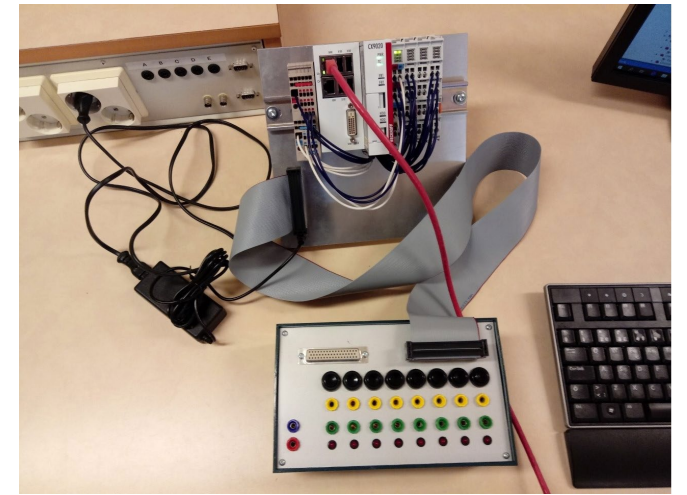
Introduction to TwinCAT

- The first programming tasks I/O-box from CANVAS
- Project Navigator
- POU Programming Organisation Unit
- Header and Addressing of physical in and outputs
- Function Block Diagram (FBD)

TwinCAT

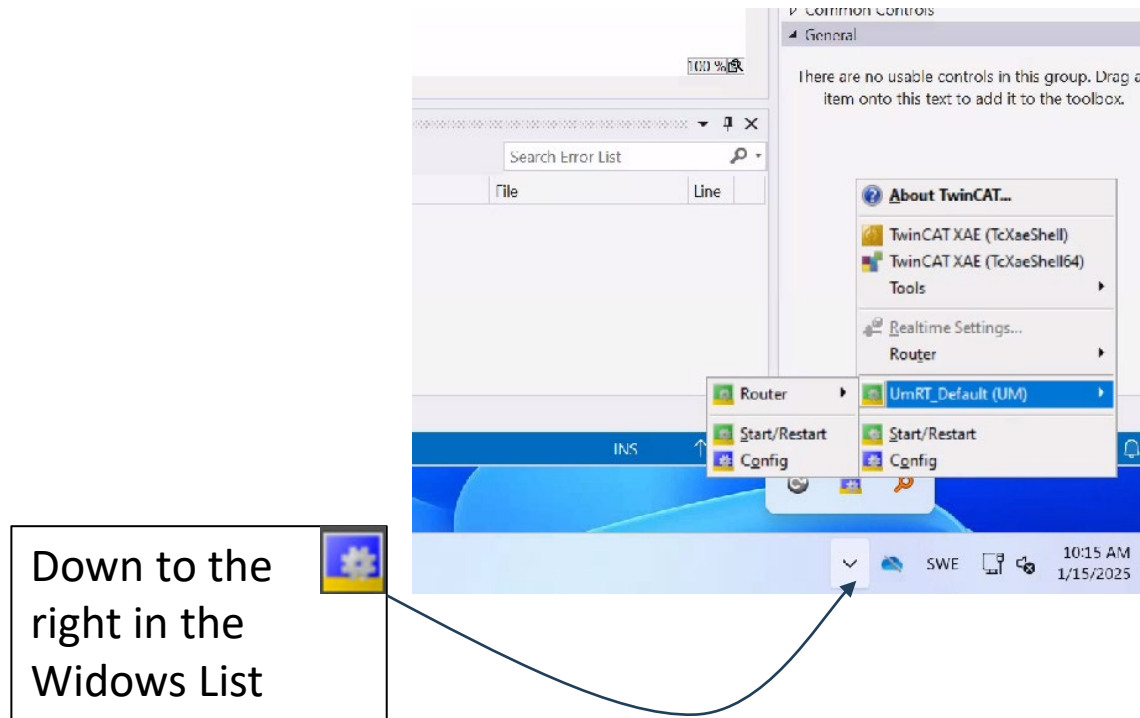
- TwinCAT is a Windows Software implemented in Microsoft Visual Studio
- All computer rooms at Campus have TwinCAT installed
- TwinCAT is reachable through RemotDesktop
- Possible to install TwinCAT on your Windows computer

I/O-Box simulation I TwinCAT

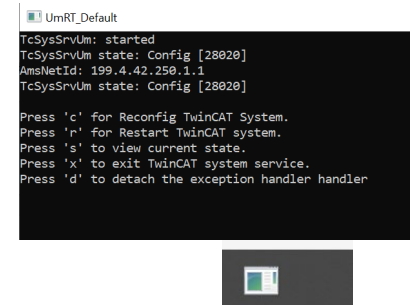


- TwinCAT Project IO Box.sln is a simulation of a box with eight buttons and eight LED lamps.
- Download TwinCAT Project IO Box from Canvas
- Extract the downloaded zip file

Launch UmRT_Default runtime PLC

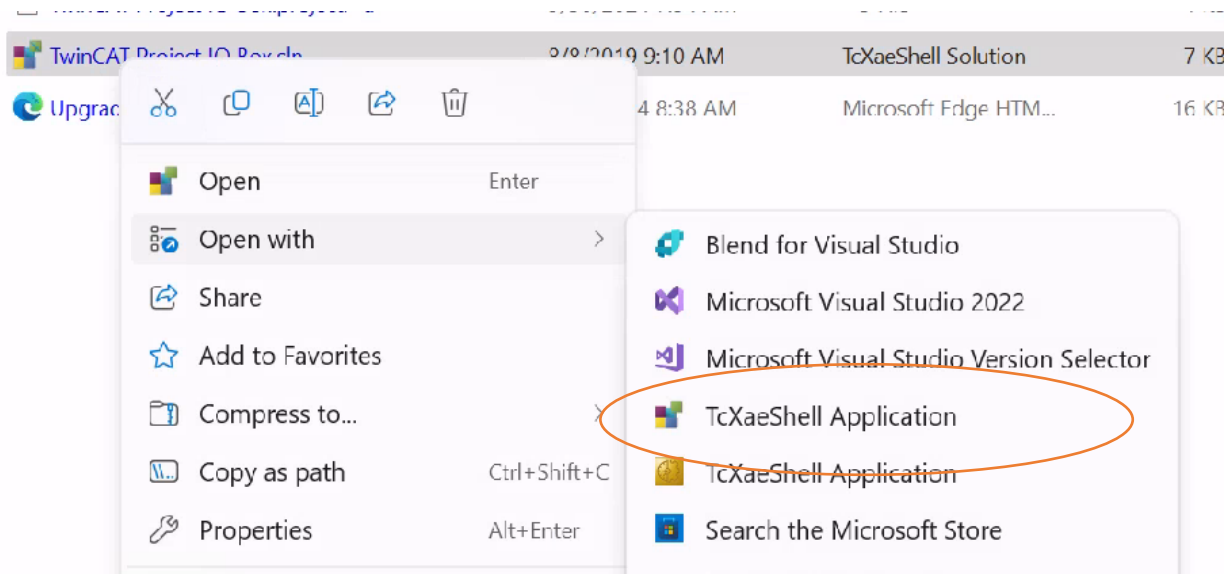


Select UmRT_Default(UM)
and click Start/Restart

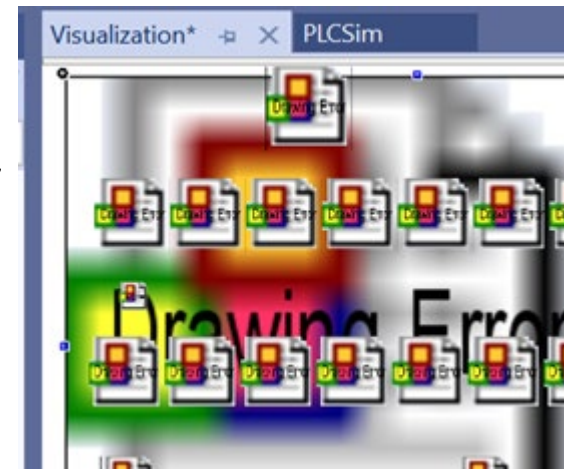
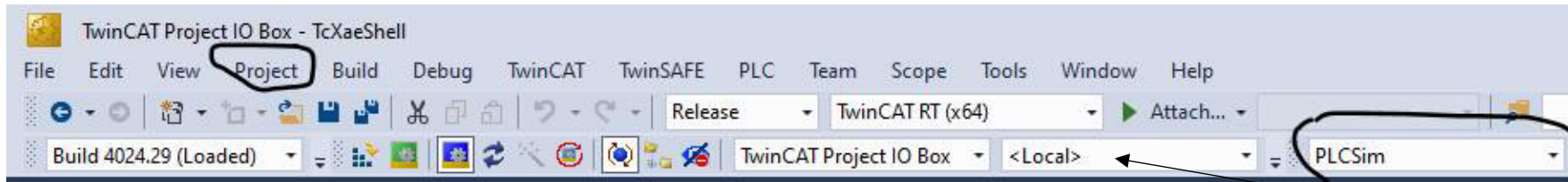


Open it with the 64-bits TcXaeShell Application

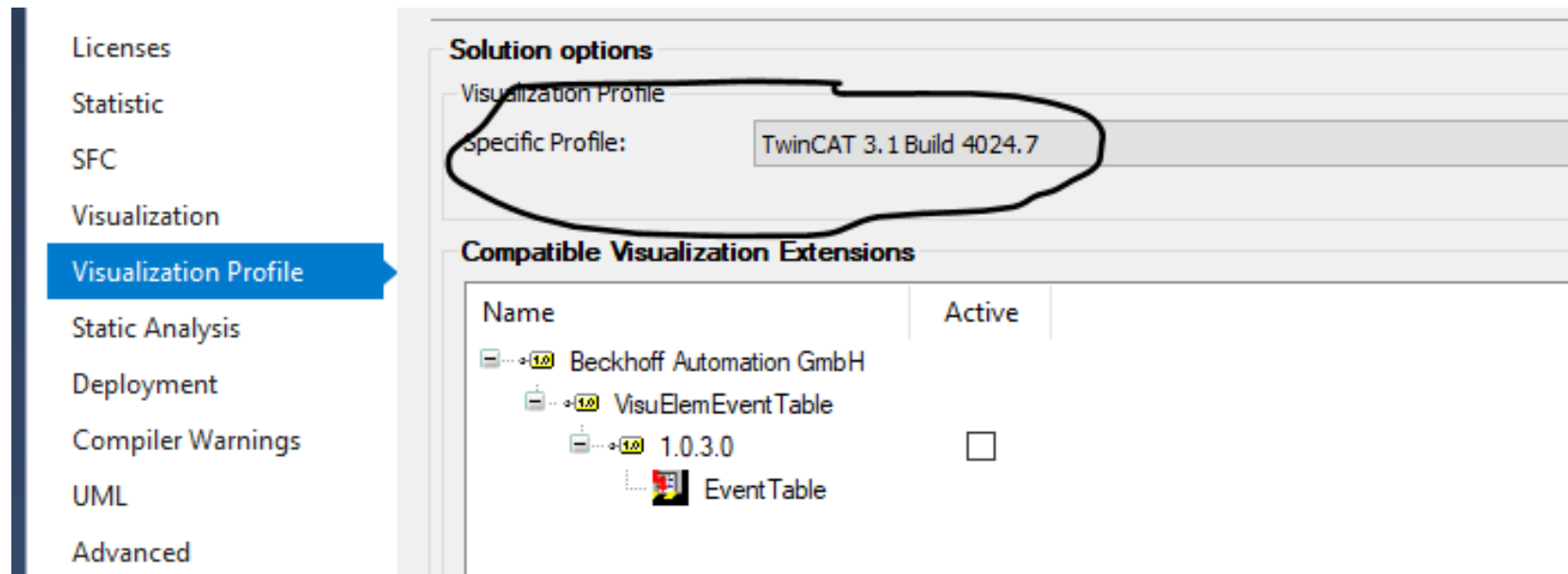
Launch the project by right-clicking on “TwinCAT Project IO Box.sln”



Select Visualization profile if necessarily

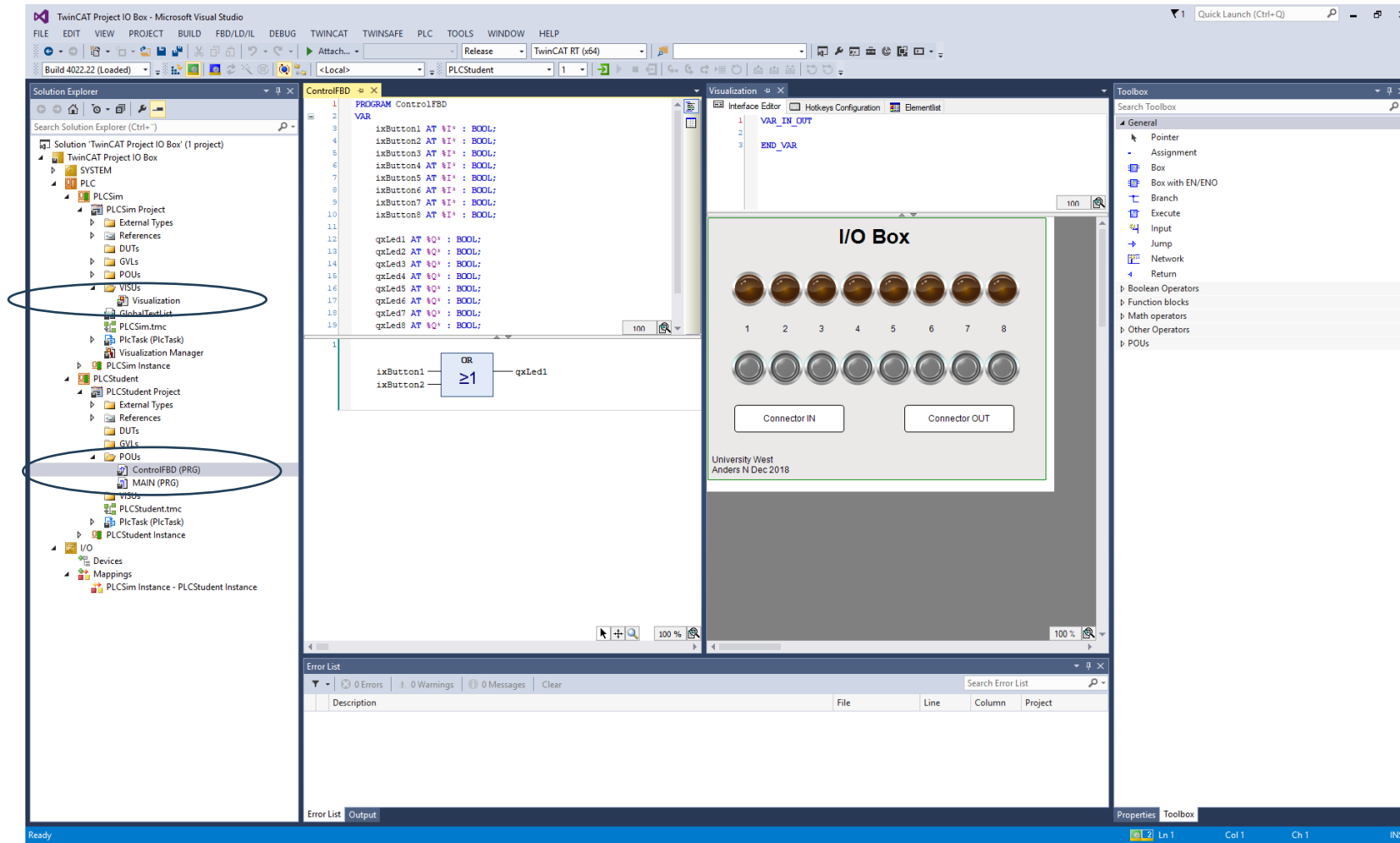


Local or UmRtDefault
depending on your
installation



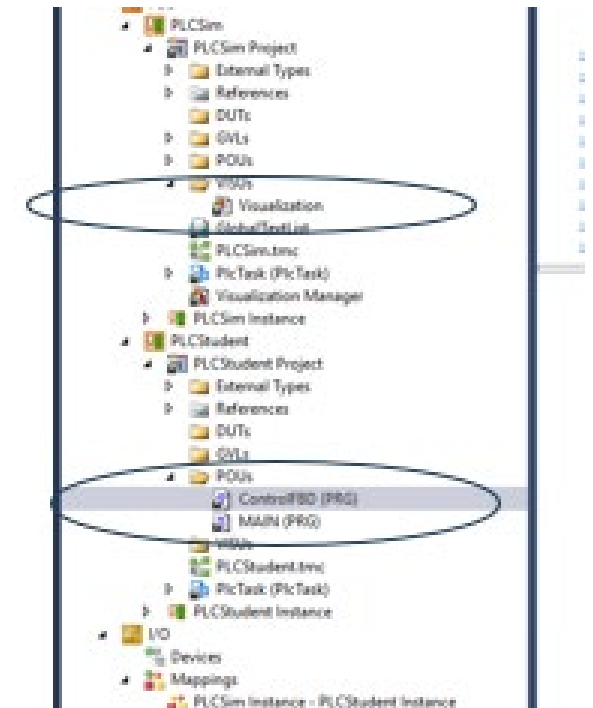
When TwinCAT launched it should look like this

If not, click in Solution Explorer on the left to bring up ControlFBD and Visualization

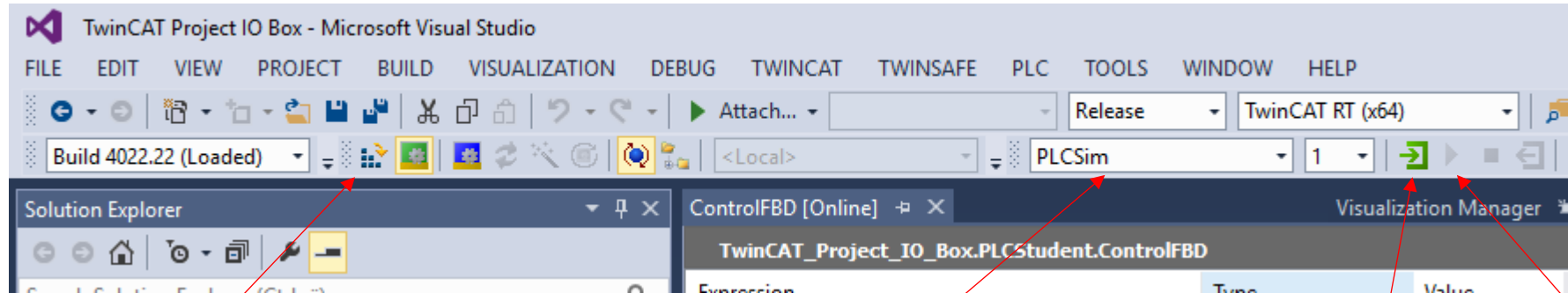


Two PLCs in one project Simulation and Student

- The project is divided into two virtual PLCs
- "PLCStudent" where you develop your programs
 - You work/programing in POU "ControlFBD"
- "PLCSim" where the simulation is located
 - You can see the simulation in VISU "Visualization"
- POU (Program Organization Unit) where the application programs in LD, FBD, ST, or SFC are located
- VISU (Visualization) a graphical interface



Run the simulation (PLCSim) and Your application (PLCStudent)



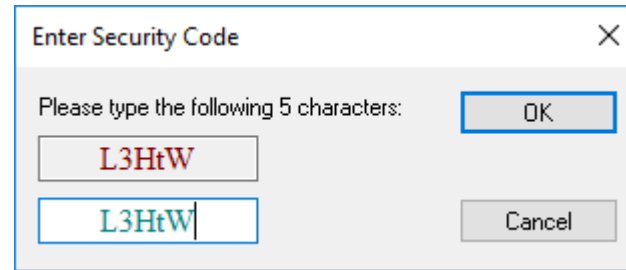
Activate Configuration
Update the actual
hardware configuration.
Needed the first time you
run the project

Selection of "PLCSim"
or "PLCStudent"

Login on the
PLC

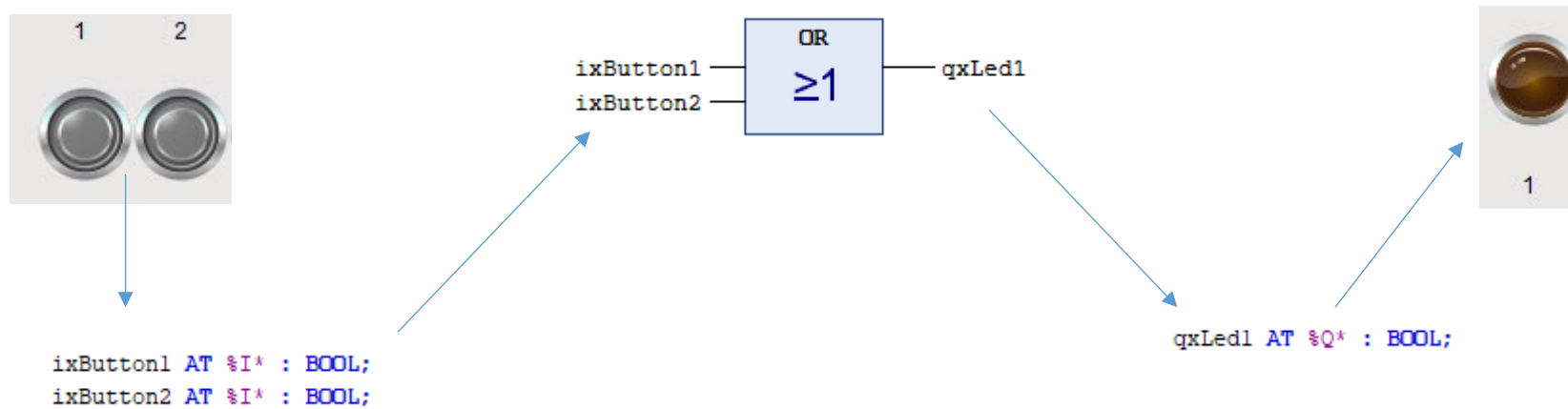
Run the
application

Generate a trial license, if asked



- Enter the suggested numbers
- Click OK
- After seven days, a new license must be generated

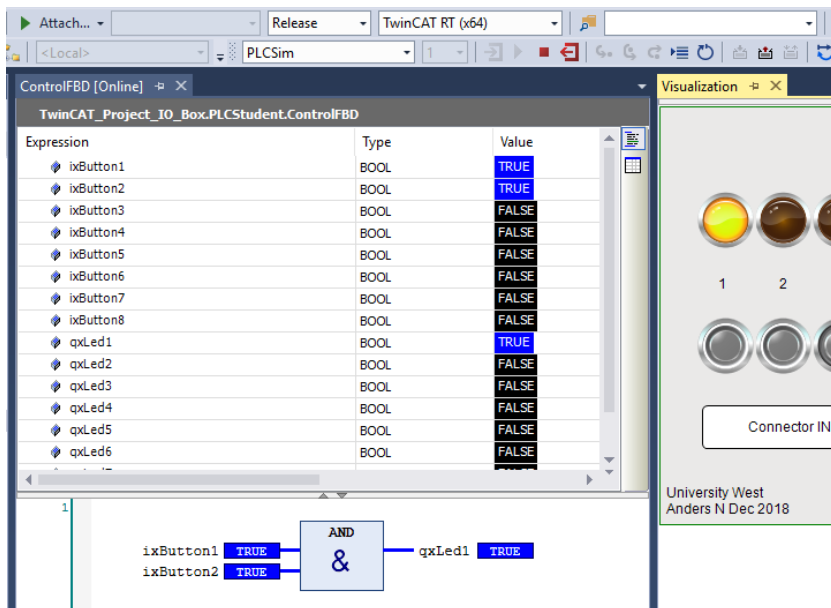
Run the project I/O-Box (Default application)



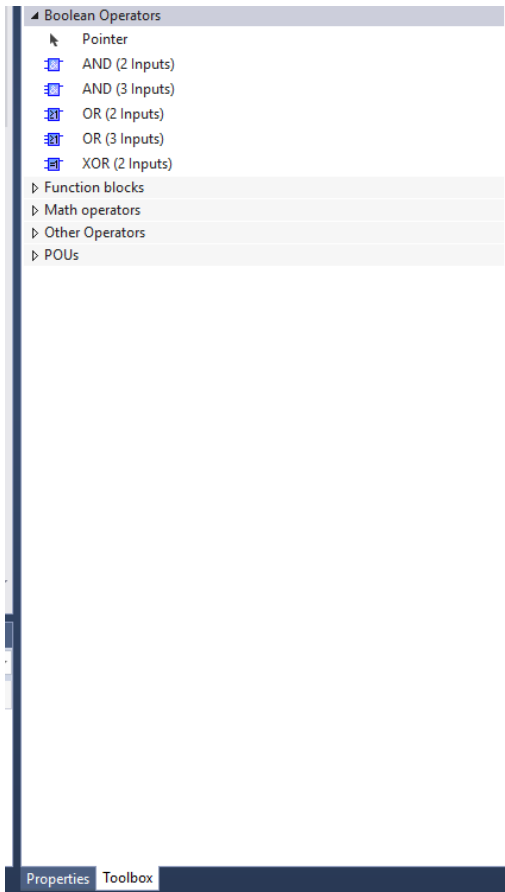
- LED1 will illuminate if button1 OR button2 is engaged
- The buttons are connected to inputs (%I*) via the variables ixButton1 and ixButton2
- The inputs are run through an OR function
- The result is assigned to the variable qxLed1, which is linked to the lamp output

Change from OR to AND

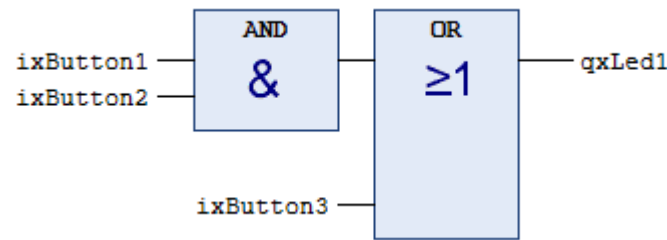
- Change the application so that Button1 **AND** Button2 must be pressed for the lamp to illuminate
- Log out of “Online Mode” click the red arrow
- Change OR to AND
 - Click on the text OR in the function and type AND
- Click on the green arrow and OK for "login with online change"



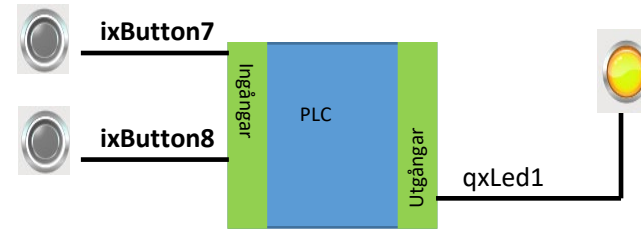
Button1 AND Button2 OR Button3



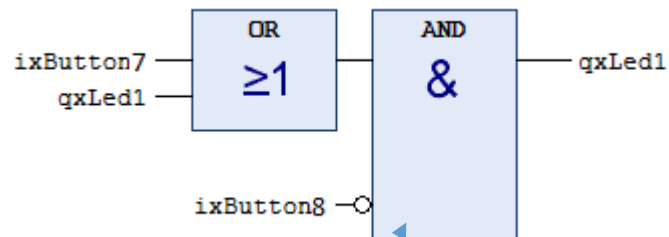
- Click on "Boolean Operation" in the "Toolbox" tab
- Drag in an OR box
- Release on the output from the AND box
 - It should light up a small green arrow where you can drop the block, drag the block to the right place if it gets wrong.
- Drag the header or enter ixButton3



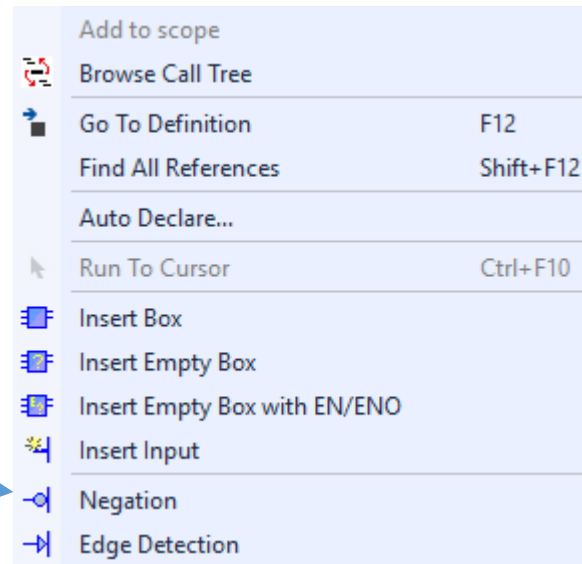
Self holding



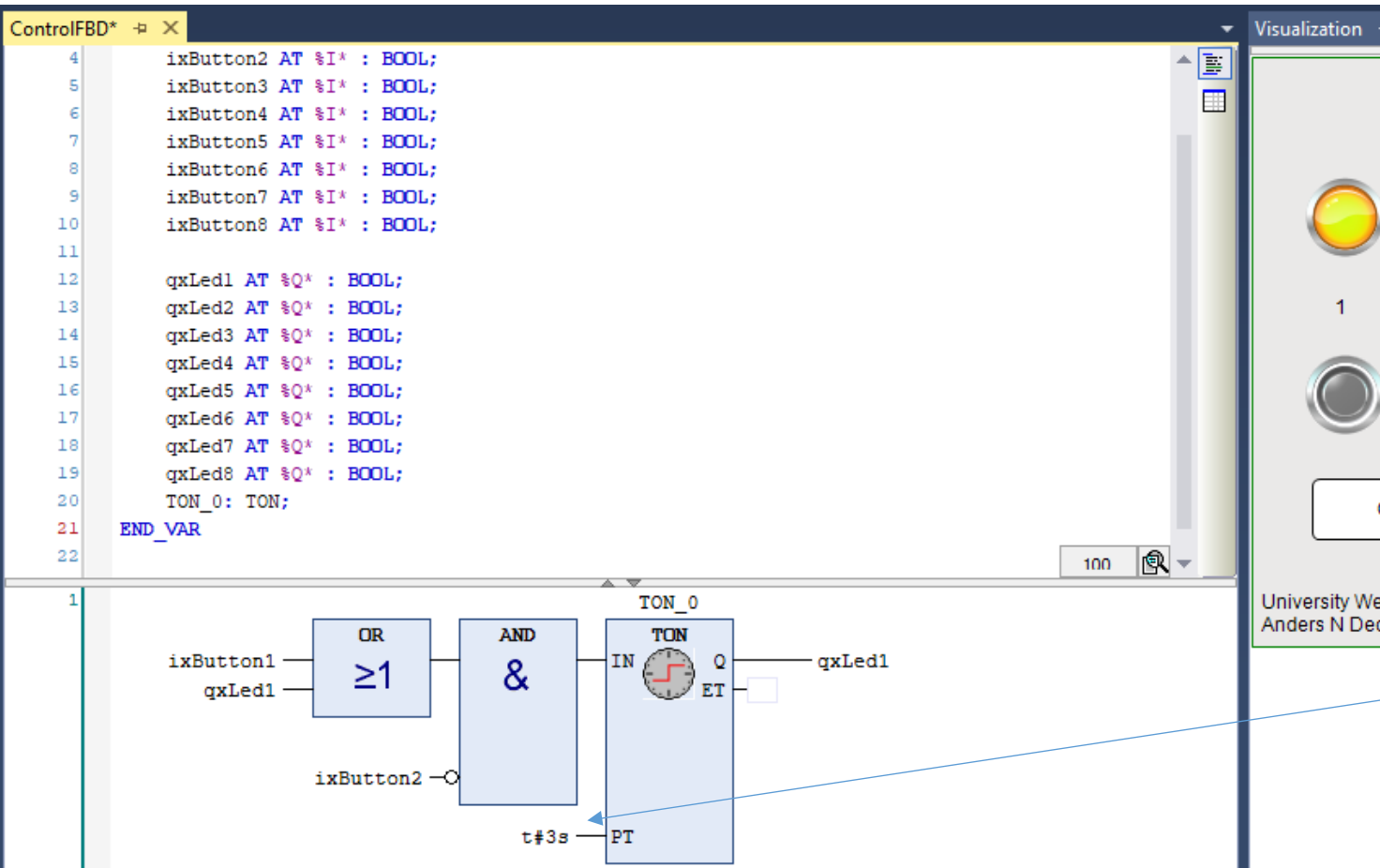
- Make an application that lights the lamp with button7 and extinguishes the lamp with button8



Right-click and Select
"Negation" to get the
inversion ring



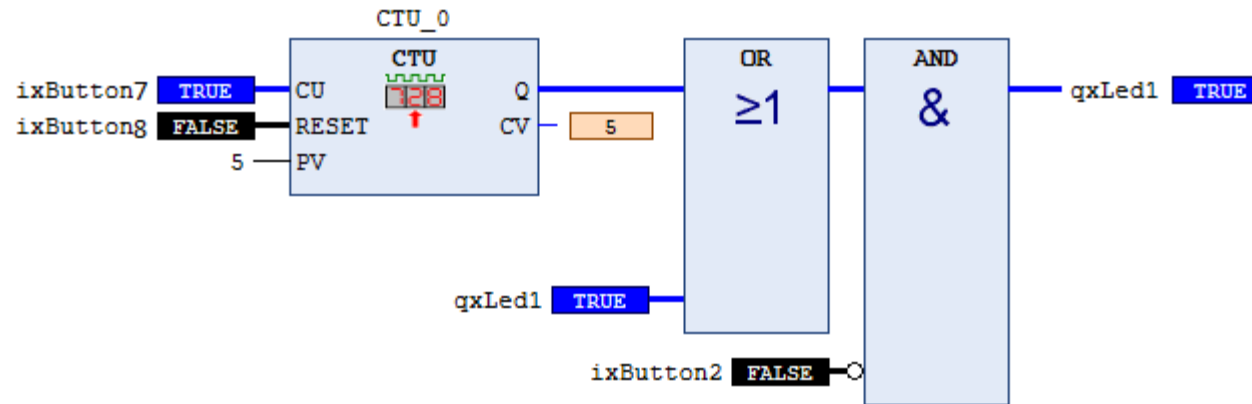
Timer Function (TON)



- Delay that turns the lamp on after three seconds
- Timers (TON) can be found in the toolbox “Function blocks”

t # 3s indicates that the value is of timer type, hours, seconds and tenths of seconds.

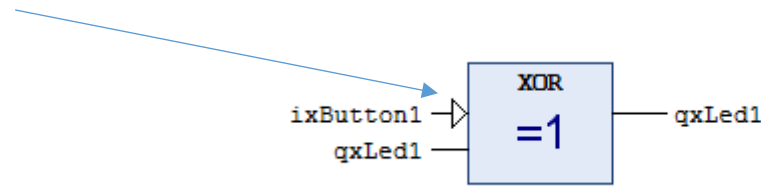
Counter (CTU)



- An application that lights the lamp after 5 press on Button7
The counter is reset with button8
- Button 2 will reset the self-holding

Toggle function, Edge detection and XOR

Right click and
select edge
detection



Switches on the lamp when the Button turns from FALSE to TRUE and un light next time